WEEKS 4 & 5: QUANTUM PHYSICS
INTERPRETATIONS
Weeks 4 & 5 • Fall 2025

1. INTRODUCTION: WHY SHOULD WE BELIEVE THIS?

1.1 The Probabilistic Nature of Reality

A radioactive atom doesn't decay on a fixed schedule. Instead, it has a probability of decaying within a certain time. We can't predict exactly when a single atom will decay—only the probability over time. This randomness is why quantum mechanics isn't just an incomplete theory; the probabilistic nature is baked into reality itself.

But maybe this doesn't make sense! Perhaps we're just missing information. Let's explore this idea...

2. MAYBE PROBABILITIES MAKE NO SENSE!

2.1 Hidden Variables

The Idea: What if we just don't know certain properties of particles yet?

- If we measure an electron's spin as üp,"we say it could have also been spin "down"
- A hidden variable theory would say it was already up—it just looked random because we didn't know the hidden details
- This would restore determinism: particles have definite properties, we just can't see them all

Disproving This: The Problem of Entanglement

Imagine we have two electrons that are **entangled** (their spins are correlated):

- If we measure one electron's spin and find it's üp,"then the other must be "down"
- This correlation holds even if the electrons are light-years apart
- When we measure one, the other *instantly* knows what to be

This creates two deeply uncomfortable possibilities:

1. Some kind of influence travels faster than light (violating relativity)
2. Reality is fundamentally **non-local**, meaning measurements on one particle instantly affect the other, even across vast distances

Bell's Theorem (1964) proved that no local hidden variable theory can reproduce the predictions of quantum mechanics. Experiments have confirmed quantum mechanics is correct, which means either:

- The universe is non-local (spooky action at a distance is real), or
- Reality doesn't exist until we measure it (there are no hidden variables)

Both options are profoundly unsettling!

3. Quantum Effects Beyond the Lab

3.1 Black Holes and "Virtual"Particles

We learned that transistors and memory storage rely on quantum effects. But quantum mechanics governs much more than just technology—it even explains how black holes behave! **Hawking Radiation: Black Holes Aren't Perfectly Black**
In 1974, Stephen Hawking made a calculation that revolutionized our understanding of black holes. He showed that black holes aren't perfectly black—they emit radiation and slowly evaporate. This was revolutionary because the defining feature of a black hole is that nothing can escape, especially not light. Yet quantum mechanics says they're actually glowing, just very, very faintly.

The Quantum Vacuum: Space Isn't Empty

- In quantum mechanics, space is not actually empty
- Quantum fluctuations cause pairs of "virtual particles"to pop in and out of existence, even in a vacuum like empty space!
- Normally, these pairs annihilate each other almost instantly, like an electron and a positron meeting
- Heisenberg uncertainty is cool with this!

Why the Uncertainty Principle Allows This:

For a very short time (Δt), the uncertainty in energy (ΔE) can be large:

$$\Delta E \cdot \Delta t \geqslant \frac{\hbar}{2}$$

This allows "borrowed.energy to spontaneously create a particle-antiparticle pair, as long as they annihilate quickly enough to return"the borrowed energy.

How Black Holes Steal Particles

Near a black hole's event horizon (the point of no return), the gravitational field is so extreme that it creates a sharp boundary in spacetime. Think of this like a waterfall—a sharp drop where if you fall in, you're done for!

Sometimes, a virtual particle pair appears right at this edge:

1. The intense gravity can separate them before they annihilate
2. One particle falls into the black hole
3. The other escapes as real Hawking radiation

We know $E = mc^2$, so when one particle escapes, it carries away energy. Therefore, the black hole loses mass. The idea is that eventually (after a longer time than is particularly meaningful), all black holes will evaporate into nothing!

An Analogy: The Cosmic Bank Account

Think of it like a cosmic bank account: the universe briefly "loans.energy to create particles. Usually, they pay it back immediately by annihilating. But the black hole intervenes, keeping one particle while letting the other go free. By keeping one, the black hole has stolen from the bank and now has a debt that shrinks the black hole's bank account (mass).

Why Haven't We Detected It?

No one has detected Hawking radiation because all known black holes are far too massive and cold. Stellar-mass black holes are actually absorbing more energy from cosmic microwave background radiation than they emit as Hawking radiation—they're growing, not shrinking. Only after the universe expands and cools for unimaginable eons would these black holes begin to evaporate.

4. Big Problems (Uh-Oh)

4.1 Unresolved Mysteries in Physics

This all might seem pretty believable—the math certainly backs it up. But why don't physicists claim they understand the universe yet? That's because we don't, and there are significant issues with our theories that we don't have the answers to yet.

Gravity and the Four Forces

- Classically, we think of gravity as the curvature of spacetime (General Relativity)
- Quantum mechanically, we like everything to be "quantized": everything is a wave that can act like a discrete particle
- The other three forces (electromagnetic, strong nuclear, weak nuclear) all have force-carrying particles
- There should therefore be a particle that carries the gravitational force

Gravity's "Force Carrier"Should Be the Graviton

But we've never detected a graviton, and when we try to include it in quantum equations, we get infinite energy values that can't be canceled out. General Relativity predicts singularities inside black holes and at the Big Bang—places where spacetime curvature becomes infinite. These infinite values mess up the math!

5. Discussion Questions

5.1 Hidden Variables and Measurement

1. If hidden variables don't exist, does this mean the act of measurement creates reality?

2. What would it mean for reality if quantum mechanics turns out to be non-local?

3. Classical physics suggests a deterministic universe—everything follows predictable laws. Quantum mechanics suggests randomness is fundamental—things don't have predetermined outcomes. If the universe is fundamentally random, does that mean free will exists, or are we just ruled by probabilities instead of classical determinism?

a. Does randomness give us freedom, or does it just mean we lack control over anything?

5.2 Black Holes and the Nature of Reality

1. Does the idea of empty space being full of energy change how you think about reality?

2. If particles can spontaneously appear, could the entire universe have been created this way?

3. The "black hole information paradox"suggests that information about particles falling into a black hole might be lost forever. This contradicts quantum mechanics, which says information can't be destroyed. If black holes destroy information, does that mean reality itself can be erased?

a. If information can truly disappear, does that mean reality is not as permanent as we think?

5.3 Problems with Gravity and Fundamental Limits

1. Could spacetime itself be quantized, meaning space and time come in discrete chunks rather than being continuous?

2. If quantum mechanics and general relativity contradict each other, does that mean we will never fully understand reality?

a. If they can't be unified, is there a fundamental limit to human knowledge? Will we ever understand 100 % of our universe?

5.4 Interpretations: A Personal Reflection

In your group, open your homework essays about quantum mechanics interpretations.

a. Did you all pick the same interpretation? Why or why don't you agree?

b. Has your opinion on a particular interpretation changed over time?

c. Share a line or two from your essay with each other, specifically an *opinion* you wrote. Comment your thoughts (in a line or two) on each person's perspective.

Reflection: There is no çorrectïnterpretation of quantum mechanics. Each interpretation makes the same experimental predictions, but they differ in what they say about the nature of reality. Your choice reflects your philosophical intuitions about what seems most reasonable or least objectionable. As you learn more physics and philosophy, your views may evolve—and that's part of the journey of understanding the quantum world!